Separation and reunion of a gorilla
Gorilla g. gorilla
infant and mother

RONALD D. NADLER[1] & SUSAN GREEN[2]
[1]*Developmental Biologist, Yerkes Regional Primate Research Center, Emory University, Atlanta, Georgia 30322, USA and* [2]*5424 Arlington Avenue, Riverdale, New York 10471, USA*

On 15 June 1973 a nine-month-old ♀ Lowland gorilla *Gorilla g. gorilla* 'Patty Cake', the first gorilla born in New York City, was reunited with its mother following a three month period of separation. The significance of the event lies in the fact that separation for a prolonged interval and reunion took place while the infant was less than one year old, a stage of development in gorillas that is characterized by a strong dependency on the mother (*10*). It is noteworthy both because infant gorillas are frequently separated from mothers early in life for a variety of reasons (*8*) and because such separation in apes may adversely influence adult social behaviour and reproduction (*3; 7*).

BACKGROUND

The parents 'Lulu' and 'Kongo' arrived in the Central Park Zoo on 11 May 1966. On arrival Lulu weighed 8·6 kg and Kongo 10 kg. They were both estimated to be 13–14 months old. The pair lived together continuously from the time of their arrival and after 1971 were observed to engage in copulation frequently. However, the birth of Patty Cake came as a complete surprise because it was thought that Lulu was still too young to conceive.

The birth took place at 1610 hours on 3 September 1972 in an outdoor exhibition cage, with the ♂ present, and in full view of a group of visitors. The delivery was rapid and by the time zoo keepers arrived Lulu was already holding the infant on her ventrum. Kongo showed considerable interest in the new arrival but remained at a distance and eventually left the cage without outside intervention. Because Lulu showed signs of taking care of the infant and because he recognized the advantages of maternal rearing the Director, John W. Fitzgerald, allowed Lulu to retain the infant. However, the ♂ was not allowed to rejoin them.

Although generally a conscientious mother, Lulu exhibited temporary periods of indifference during the first month when she left the infant screaming on the cage floor. It was therefore decided to reintroduce Kongo. This proved to be a successful strategy and the trio remained together compatibly until 20 March when Patty suffered an injury. The ♀ and infant had been separated temporarily from the ♂ in an adjoining cage to provide them with some respite from his activities. Unfortunately no zoo personnel were present when the injury occurred. However, visitors stated that Patty Cake had reached through the bars to the ♂ when suddenly Lulu pulled the infant to her while Patty's arm was trapped between the bars. Paediatricians from the New York Medical College diagnosed a broken arm which was set and a cast applied.

SEPARATION

Patty Cake was placed in the care of Dr Emil P. Dolensek at the Bronx Zoo Hospital Nursery, since the Central Park Zoo lacked the facilities for such care. He replaced the cast with a sling which remained in position for three weeks.

Dr Dolensek found that Patty weighed less than nursery-reared gorillas of the same age, her central incisors were discoloured and the enamel appeared thin; faecal examination disclosed the presence of *Balantidium coli* and strongyle ova and a nasal swab contained *Staphylococcus aureus* and *Escherichia coli*. Thiabenzole (50 mg/kg) was administered as a single injection for the strongyles and tetracycline (50 mg/kg) was given for seven days to treat the *B. coli*. In addition, DPT vaccine and Salk's oral polio vaccine were each given twice: the day of arrival and a month later.

Patty's recuperation progressed rapidly such that she regained partial use of her arm by the time the sling was removed and full use by six weeks. Her weight increased sharply over that time from 3·7–5·8 kg. The question then arose whether it was appropriate to return her to the Central Park Zoo or to hand-raise her with two young ♂ gorillas of similar age in the Bronx Zoo Nursery – 'Mopi' born 29 May 1972 at the National Zoo Washington and 'Hodari' born 2 October 1972 at the Bronx Zoo. Richard M. Clurman (Commissioner of Parks, Recreation and Cultural Affairs for the City of New York) and William G. Conway (General Director of the New York Zoological Society) agreed that the primary consideration was that the rearing conditions give Patty Cake the best opportunity to develop into a socially competent gorilla, capable of breeding and rearing offspring of her own.

One of the authors (RDN) was asked to investigate the relevant factors. Review of the literature suggested that of the two alternatives, returning Patty to her parents offered the best chance of her breeding successfully in the future. Research conducted on chimpanzees at the Yerkes Regional Primate Research Center showed that nursery-reared chimpanzees generally weighed more than mother-reared animals of the same age and, indeed, exhibited superior physical health (*2*). However, the nursery-reared animals developed into social (*9*), psychological (*1*) and intellectual (*4*) misfits, and participated in copulatory activity only when tutored and/or coerced by wild-born partners. Furthermore, none of the ♀♀ ever reared an infant and frequently infants which were not immediately removed were found dismembered or otherwise seriously abused. Also noteworthy was the suggestion that the detrimental effects of nursery-rearing were even greater in animals given extensive human contact than in those raised in nearly complete social isolation (*3; 9*).

Predictions from laboratory experiments to situations outside the laboratory vary in value. However, the consistent development of conspicuous behavioural anomalies in nursery-reared animals so closely related to the gorilla was highly suggestive and, in light of the seriousness of the resultant condition, received predominant consideration. Other factors that were considered related specifically to the animals and the environment in question and included the previous behaviour of the parents, especially the mother, with the infant, the circumstances under which the fracture occurred, the physical health of the infant at the time of the accident, and the physical environment at the Central Park Zoo.

Prior to the injury one of the authors (SG) had kept a record of the gorillas' behaviour on a 7–9 hr daily schedule, which indicated that parents and infant had enjoyed an amicable and congenial relationship. Lulu had been a possessive but effective mother and Kongo a considerate, though somewhat distant father (*5*), an impression confirmed by the Zoo staff. Moreover, reports from visitors who had witnessed the event seemed to agree that the arm was broken accidentally.

Although the medical report on Patty Cake raised questions about her health and the possibility of malnutrition, con-

sideration of the individual symptoms suggested that a more optimistic view was possible. Patty's relatively low body weight at the time of separation and the sharp increase during two months of bottle feeding appear to be characteristic of apes subjected to such rearing conditions and not indicative of a pathological condition (*2*). Similarly, bacterial and parasitic conditions are common problems in captive apes, and while undesirable, are generally tolerated by the animals as long as routine examinations and treatment are provided (*6*).

The section of closely spaced bars that contributed to Patty's fracture was modified following the accident and the housing, although old and due for replacement, was considered adequate in all other respects.

It was therefore decided to attempt to reunite Patty and her parents. As a preliminary step Patty's original keepers at Central Park Zoo, Eldemiro Rodriguez and Veronica Nelson, visited Bronx Zoo daily for one week to feed and handle Patty, who quickly readjusted to them. When she was returned to Central Park Zoo, she was not immediately introduced to Lulu and Kongo but was allowed for one week to explore the cages alone, for increasing periods each day.

REUNION

With the building closed to the public and the adult gorillas locked outside, Patty was placed in one of the indoor cages, and Lulu allowed into an adjoining cage. Originally it was decided to observe the animals' reactions for a time while they remained in separate cages. However, as Lulu entered the cage adjoining Patty's, she emitted a series of loud cries and became highly agitated. Patty appeared startled and did not approach her mother. In order to avoid upsetting the animals further, the door between the two cages was opened sooner than planned. Lulu quickly entered Patty's cage, walked toward the infant and reached down as if to touch her or pick her up. The infant responded by screeching and cowering and Lulu withdrew. This pattern was repeated several times and Patty's resistance appeared to diminish with each successive encounter. After approximately five minutes Lulu grasped Patty with her right hand and the infant wrapped both its arms around the mother's arm. Lulu raised Patty to her lower ventrum, sat down and gradually began to examine her newly returned offspring.

After about 20 minutes the ♂ was allowed into the adjoining cage and in another few minutes into the same cage, but he did not approach the pair. After a short time, however, Lulu carried Patty to him and allowed him to touch, smell and examine the infant as he had done in the past. It was apparent that the first phase of the reunion was successfully accomplished.

One problem that remained was the supplementary feeding of Patty Cake. It was assumed that Lulu had ceased lactating during the three months' separation and it was feared that Patty could not compete adequately with the adults for food. Therefore, it was decided to continue feeding Patty the infant formula she had received at the Bronx Zoo. Lulu and Patty were coaxed by their keepers to approach the front of the cage and since Lulu was as interested in the formula as Patty, both animals were given bottles to assure the infant an uninterrupted meal. The supplementary feeding and routine health examination represent a continuing effort to ensure the maintenance of good health and appear to date to have been successful.

A few weeks after the reunion Lulu resumed lactation, although following her return Patty was never observed to suckle. Apparently, the presence of, and the contact with the infant were sufficiently stimulating to reinitiate lactation, even in the absence of actual suckling! Lacking further stimulation, however, lactation was not maintained and Lulu began to exhibit cyclic displays of

mating activity. During those periods she neglected Patty, who for her own safety was separated from the adults each month for the two or three day oestrous period. This plan has been successfully applied at other times when the aggressiveness of the adults threatened the well-being of the infant. During such periods of separation, Patty shared a cage with a young chimpanzee, so as to provide her with some nonhuman companionship and diversion.

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PRODUCT MENTIONED IN THE TEXT

Thiabenzole^{R}: manufactured by Merck & Co., Rahway, New Jersey 07065, USA.

REFERENCES

1. DAVENPORT, R. K. Jr & MENZEL, E. W. Jr (1963): Stereotyped behavior of the infant chimpanzee. *Arch. gen. Psychiat.* **8:** 99–104.

2. DAVENPORT, R. K. Jr, MENZEL, E. W. Jr & ROGERS, C. M. (1961): Maternal care during infancy: its effect on weight gain and mortality in the chimpanzee. *Am. J. Orthopsychiat.* **31:** 803–809.

3. DAVENPORT, R. K. & ROGERS, C. M. (1970): Differential rearing of the chimpanzee. A project survey. In *The chimpanzee* **3:** 337–360. Bourne, G. H. (ed.). Basel: S. Karger.

4. DAVENPORT, R. K., ROGERS, C. M. & RUMBAUGH, D. M. (1973): Long-term cognitive deficits in chimpanzees associated with early impoverished rearing. *Devl. Psychol.* **9:** 343–347.

5. GREEN, S. (In press): *Patty Cake, metropolitan gorilla: the first year in the life of a captive gorilla.* New York: Simon and Schuster.

6. McCLURE, H. M. & GILLOUD, N. B. (1971): Comparative pathology of the chimpanzee. In *The chimpanzee* **4:** 103–272. Bourne, G. H. (ed.). Basel: S. Karger.

7. MITCHELL, G. (1970): Abnormal behavior in primates. In *Primate behaviour:* 195–249. Rosenblum, L. A. (ed.). New York: Academic Press.

8. NADLER, R. D. (1974): Periparturitional behavior of a primiparous lowland gorilla. *Primates* **15:** 55–73.

9. ROGERS, C. M. & DAVENPORT, R. K. (1969): Effects of restricted rearing on sexual behavior of chimpanzees. *Devl. Psychol.* **1:** 200–204.

10. SCHALLER, G. B. (1963): *The mountain gorilla, ecology and behavior.* Chicago: Chicago University Press.

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The social behaviour of capybaras *Hydrochoerus hydrochaeris* at Evansville Zoo

SUSAN L. DONALDSON, TIMOTHY B. WIRTZ & ALLEN E. HITE
Department of Psychology, Indiana State University, Evansville Campus, Indiana, USA

Mesker Park Zoo in Evansville has successfully bred and raised capybaras *Hydrochoerus hydrochaeris* for a number of years under semi-natural conditions and so offers a unique microcosm for studying the behaviour of this little known South American rodent. The world's largest rodent is a phylogenetically ancient, semi-aquatic species. It is surprising that, apart from Walker's account (*1*), so little has been written on the social and breeding behaviour of a mammal which is not only of intrinsic interest to zoos, but also provides ethologists,